Shape-Oriented Convolution Neural Network for Point Cloud Analysis

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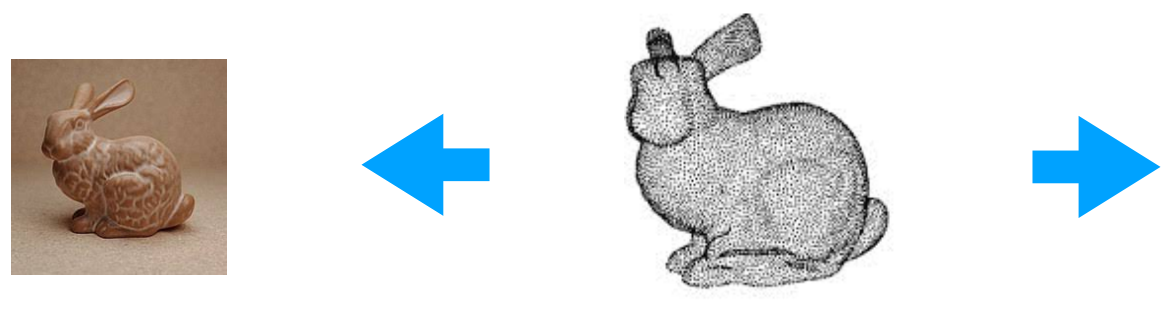
Point Cloud: a principle data structure for 3D geometric information encoding

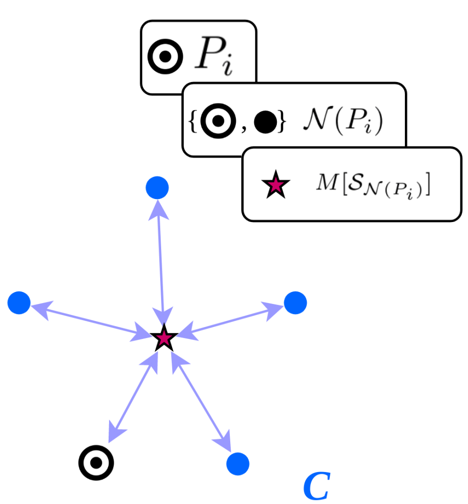
Unlike other conventional visual data (images/videos), these irregular points describe the complex **shape features of 3D objects**, which makes shape feature learning an essential component of point cloud analysis.

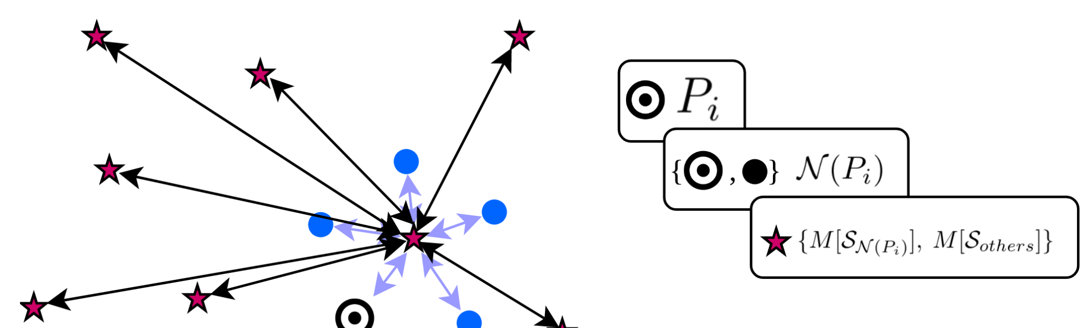


$$X_{input} \in \mathbb{R}^{N \times 3}$$

Method: shape-oriented relationships

- ***Intra-shape relationship learning* (left):** representation learning of the underlying shape formed by each local neighboring point.
- ***inter-shape relationship learning* (right):** contextual effects from the long-ranged dependencies between local underlying shapes.





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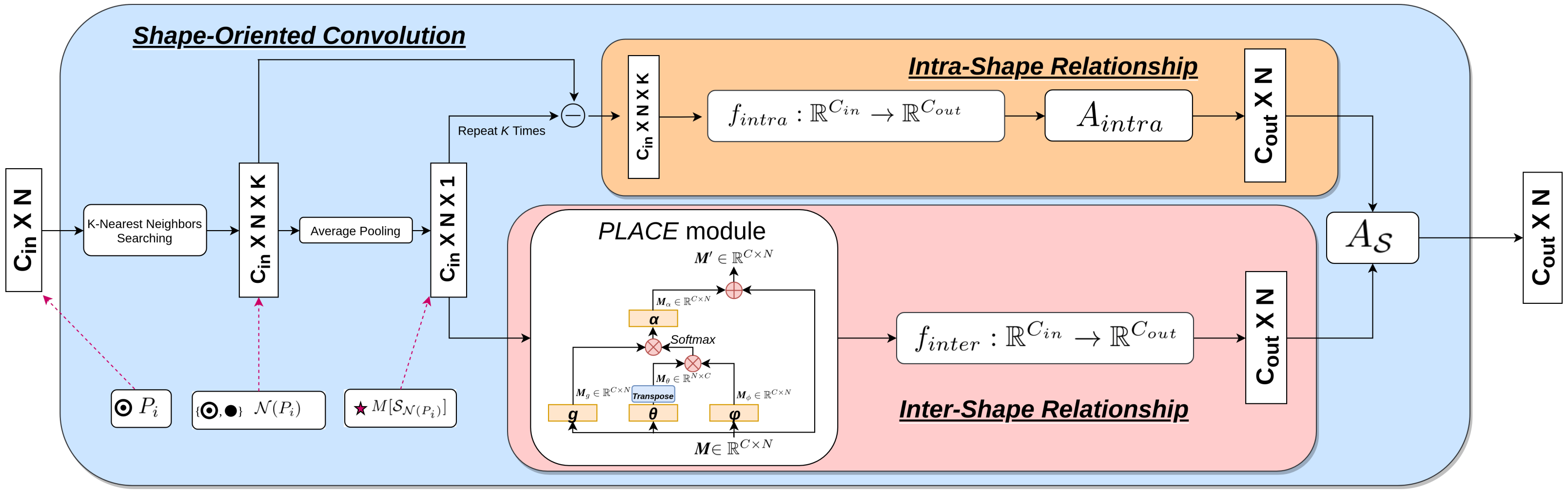
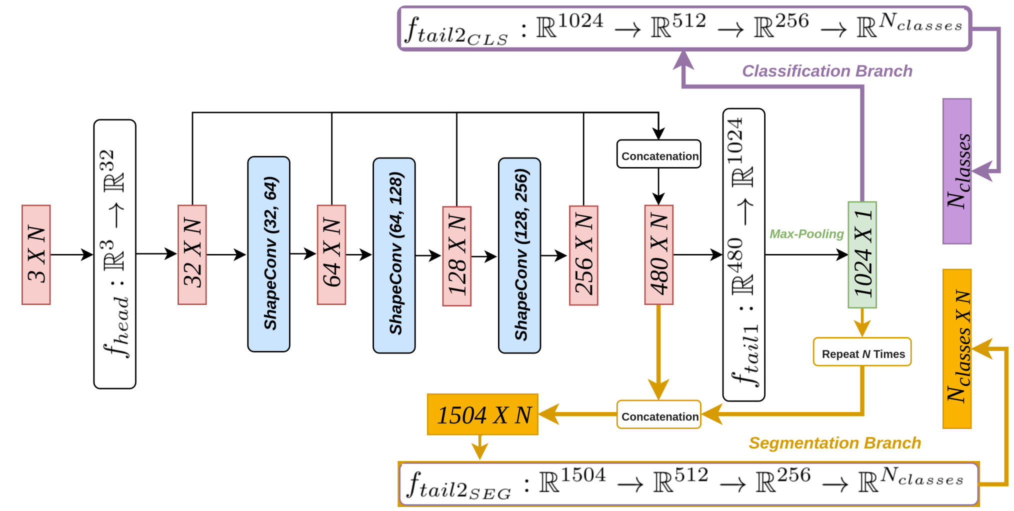
ShapeConv

(Shape-Oriented Convolution Operator)



SOCNN

(Shape-Oriented CNN)



Experimental Results

A. Point Cloud Classification (ModelNet40)

This dataset consists of 9843 training 3D models and 2468 testing models, which are collected for 40 shape categories.

Method	#points	Acc.
ECC (2017)	1k	87.4
PointNet (2017a)	1k	89.2
PointNet++ (2017b)	1k	90.7
PointNet++* (2017b)	5k	91.9
KD-Net (2017)	1k	90.6
KD-Net* (2017)	5k	91.8
SpiderCNN* (2018)	5k	92.4
SO-Net* (2018)	2k	90.9
PCNN (2018)	1k	92.3
DGCNN (2019)	1k	92.2
DGCNN* (2019)	2k	93.5
PointWeb (2019)	1k	92.3
RS-CNN (2019)	1k	93.6
RS-CNN* (2019)	2k	93.6
Proposed	1k	93.1
Proposed*	2k	93.6

B. Point Cloud Segmentation (ShapeNet-Part)

This dataset consists of 16881 3D objects, covering 16 shape categories, with 50 kinds of part-annotations in total.

Our model achieves state-of-the-art performance on the part segmentation task of the *CHAIR* objects and the *LAPTOP* objects.